

## Tutorial-8

Q1: Let  $A \in M_{2 \times 2}(\mathbb{C})$  be a rank one matrix.  
and  $\text{Tr}(A) \neq 0$

Show that  $A$  is diagonalizable.

Sol: Since, the rank  $(A) = 1$   
then  $\det(A) = 0$

for  $(2 \times 2)$  matrix then  $\text{Char}(A)$

$$\text{Char}(A) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$$

$$P(\lambda) = \lambda(\lambda - \text{tr}(A))$$

eigenvalues of  $A = 0, \text{tr}(A)$

$\therefore$   $A$  has two distinct eigenvalue then  
it is diagonalizable.

Q2: Let  $T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{bmatrix}$  be a real matrix,

under what condition on  $a, b, c$  is  $T$ -diagonalizable?

Sol:  $\text{Char}(T) = \lambda^4$

eigen values of  $T$  are  $\lambda_i = 0 \quad i = 1, 2, 3, 4.$

Let  $E(\lambda, T)$  denotes the eigenspace of  $T$  corresponding to eigenvalue  $\lambda$ .

We know that  $T$  is diagonalisable iff  $\dim E(\lambda_i, T) = n_i$ , where  $n_i \in \mathbb{N}$  s.t.  $(x-\lambda)^{n_i} \mid \chi_T(x)$  but  $(x-\lambda)^{n_i+1} \nmid \chi_A(x)$ .

As here for  $\lambda_1 = 0$ ,  $n_1 = 4$ .

→ so we must have  $\dim(E(0, T)) = 4$  for  $T$  to be diagonalizable.

Let's look at the eigenvector of  $T$  for  $\lambda_1 = 0$

They are the non-zero sol. of below system

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = 0 \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}$$

$$ax = 0, by = 0, cz = 0$$

$$\therefore a = b = c = 0$$

□

Q3: Suppose  $A$  is  $2 \times 2$  real matrix which is symmetric ( $A^T = A$ ). Prove that  $A$  is similar over  $\mathbb{R}$  to diagonal matrix.

Sol<sup>n</sup>: Let  $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$

Soln: If  $b = 0$  then  $A = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$  is diagonal matrix.

Suppose  $b \neq 0$ .

$$\text{then } \text{Char}(A) = \lambda^2 - (a+d)\lambda + (ad - b^2)$$

$$\lambda = \frac{(a+d) \pm \sqrt{a^2 + d^2 + 2ad - 4ad + 4b^2}}{2}$$

$$\lambda = \frac{(a+d) \pm \sqrt{(a-d)^2 + 4b^2}}{2}$$

$$\text{As } (a-d)^2 + 4b^2 > 0$$

$\therefore A$  has two distinct roots.

Hence  $A$  is diagonalizable.

Q6: Let  $T: M_{n \times n}(F) \rightarrow M_{n \times n}(F)$  be defined by  $T(B) \rightarrow AB$  for some fixed

$A \in M_{n \times n}(F)$ . Is it true that  $T$  and  $A$  have same eigen value? Show that the minimal pol. for  $T$  is the minimal pol. of  $A$ .

Sol: Claim:  $T$  and  $A$  have same eigen value.

Let  $\lambda$  be an eigen value of  $T$

then  $\exists X \neq 0$  s.t.  $T(X) = \lambda X$   
( $X \in M_{n \times n}(F)$ )



$$\text{i.e. } AX = \lambda X$$

$$\text{As } X \neq 0 \in M_{n \times n}(\mathbb{F})$$

WLOG 1-column is non-zero. So  $X e_1 \neq 0 \in \mathbb{F}^n$

$$\therefore A(X e_1) = \lambda(X e_1)$$

$\Rightarrow \lambda$  is eigenvalue of  $A$ .

Conversely, Suppose  $\lambda$  is an eigenvalue of  $A$ .

$$\text{Then } \exists X \neq 0 \in \mathbb{F}^n \text{ s.t. } AX = \lambda X$$

$$\text{Since } X \neq 0 \quad AXX^T = \lambda XX^T$$

$$\underline{XX^T \neq 0 \in M_{n \times n}(\mathbb{F})} \quad A(XX^T) = \lambda(XX^T)$$

$$T(XX^T) = \lambda(XX^T)$$

$\Rightarrow \lambda$  is e.v. of  $T$ .

Thus claim holds.

Now, let  $p_A, p_T$  denote the minimal pol. of  $A$  and  $T$  respectively.

$$\text{Let } p_A(t) = \sum_{i=1}^n q_i t^i, \quad q_i \in \mathbb{F} \quad \forall i$$

As  $p_A$  is the pol. of  $A$ .

$$\therefore p_A(A) = 0 \quad \text{i.e. } q_0 I + q_1 A + q_2 A^2 + \dots + q_n A^n = 0$$

$$\Rightarrow q_0 I + q_1 A + q_2 A(AI) + \dots + q_n A^{n-1}(A) \stackrel{=0}{=} 0$$

$$\Rightarrow q_0(I) + q_1(TI) + q_2 T^2(I) + \dots + q_n T^{n-1}(I) = 0$$

$$\Rightarrow p_A(T)I = 0$$

For any matrix  $B$

$$\therefore P_A(T)B = P_A(T)(IB) = \underbrace{(P_A(T)(I))}_{(=0)} B = 0$$

$$\therefore P_A(T) = 0$$

$$\therefore P_T / P_A$$

$$\text{Let } P_T(t) = \sum_{j=0}^m b_j t^j, \quad b_j \neq 0 \quad \forall j$$

$$\begin{aligned} \therefore P_T(T) &= 0 \quad \text{i.e.} \quad b_0 T^0 + b_1 T^1 + \dots + b_m T^m = 0 \\ &\Rightarrow \left( \sum_{j=0}^m b_j T^j \right) (B) = 0 \quad \forall B \in M_{n \times n}(F) \end{aligned}$$

In particular  $B = I$

$$\begin{aligned} 0 \left( \sum_{j=0}^m b_j T^j \right) I &= b_0 I + b_1 A I + b_2 A^2 I + \dots + b_m A^m I = 0 \\ &= b_0 I + b_1 A + b_2 A^2 + \dots + b_m A^m = 0 \\ &= P_T(A) = 0 \end{aligned}$$

$$\therefore P_A / P_T$$

$$\text{hence } P_A = P_T$$

QED

Q4: Let  $A$  be a  $n \times n$  matrix s.t.  $A^k = 0$  for some positive integer  $k$ . Show  $A^n = 0$ .

Sol: Let  $\lambda$  be an e.v. of  $A$   
then  $Ax = \lambda x = \lambda^n x$  for  $x \neq 0 \in \mathbb{C}^n$

Observe that  $A^k x = \lambda^k x$

but  $A^k = 0$

So,  $\lambda^k x = 0$

$\lambda = 0$  ( $\because x \neq 0$ )

Thus '0' is only the eigenvalue of  $A$ .

So  $\text{Char}(A) = \chi_A(x) = x^n$

by Cayley Hamilton thm  $A^n = 0$

Q115: Let  $N$  be a  $2 \times 2$  complex matrix s.t.

$N^2 = 0$ . prove that either  $N = 0$  or  $N$  is similar

to  $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ .

Sol: if  $N = 0$ , then we are done  
if  $N \neq 0$

Let  $T: \mathbb{C}^2 \rightarrow \mathbb{C}^2$  as  $T(x) = Nx$  is a linear transformation

Let  $N = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

and  $\beta = \{e_1, e_2\}$  be a standard basis.

$$\text{then } T(e_1) = Ne_1 = \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} = a_{11}e_1 + a_{21}e_2$$

$$T(e_2) = Ne_2 = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} = a_{12}e_1 + a_{22}e_2$$

$$\therefore [T]_{\beta} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = N$$

Now as  $\mu \neq 0$ , so  $N$  has a non-zero column,

wlog say first column is non-zero

so,  $Ne_1 \neq 0$

claim  $\mathcal{D} = \{e_1, Ne_1\}$  is basis for  $\mathbb{F}^2$

proof: enough to show  $(e_1, Ne_1)$  is l.i.

Let  $\alpha_1, \alpha_2 \in \mathbb{F}$  s.t.

$$\cancel{Ne_1} \alpha_1 + \alpha_2 Ne_2 = 0$$

$$Ne_1 \alpha_1 + \alpha_2 N^2 e_2 = 0$$

$$(Ne_1) \alpha_1 = 0 \quad (\because N^2 = 0)$$

$$\underline{\alpha_1 = 0} \quad (\because Ne_1 \neq 0)$$

and we get  $\alpha_2 = 0$

$$\text{Now, } T(e_1) = Ne_1 = 0 \cdot e_1 + 1 \cdot Ne_1$$

$$T(Ne_1) = N(Ne_1) = N^2 e_1 = 0 = 0 \cdot e_1 + 0 \cdot Ne_1$$

$$\therefore [T]_{\mathcal{D}} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

As  $[T]_{\beta} \& [T]_{\mathcal{D}}$  are similar for any  $\beta$  and  $\mathcal{D}$

$$\therefore N = S \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} S^{-1}$$

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